

Sliced Wasserstein Distance for Scalable Unsupervised Domain Adaptation: MNIST $\rightarrow$ SVHN

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June 11, 2026

Abstract

Unsupervised domain adaptation (UDA) addresses the problem of transferring a classifier trained on a labeled source domain to an unlabeled target domain whose underlying data distribution differs from that of the source. Full optimal transport (OT) provides a principled measure of distributional discrepancy but scales quadratically in the number of samples, making it impractical as a per-batch training loss in deep neural networks. We implement Sliced Wasserstein Distance (SWD) and Max-Sliced Wasserstein Distance (Max-SWD) from scratch as scalable plug-in alignment losses for a shared convolutional encoder trained on the MNIST $\rightarrow$ SVHN benchmark, without access to any target labels during training. Max-SWD achieves a best target accuracy of 28.19% on the SVHN test set—a 4.92 times improvement over the unadapted baseline—at only 7% additional wall-clock overhead per epoch relative to training without any alignment, while full entropic OT via Sinkhorn peaks transiently at 27.07% before degrading to 18.48% by epoch 20, confirming that quadratic scaling and plan over-smoothing undermine its viability as a sustained training signal. A projection ablation over $L \in \{10, 50, 100, 500, 1000, 2000, 5000\}$ at embedding dimension $d = 128$ establishes $L = 1,000$ as the practical convergence threshold, where the coefficient of variation of the Monte Carlo SWD estimator drops below 5% at a wall-clock cost of 35.56 ms per call: approximately $33\times$ faster than Sinkhorn at equivalent sample size.

1 Introduction

Modern supervised learning systems depend on the assumption that training and test data are drawn from the same underlying distribution. In practice, this assumption is routinely violated: a model trained on one data source—the *source domain*—is deployed on data from a related but statistically distinct *target domain*, and performance degrades in proportion to the severity of the distributional mismatch [Ben-David et al., 2010]. This problem, known as *covariate shift* [Ben-David et al., 2010], is especially pronounced in computer vision, where differences in illumination, background clutter, image resolution, and color statistics can produce large distributional gaps even between datasets that share the same label space. The MNIST $\rightarrow$ SVHN benchmark instantiates this challenge concretely: MNIST digits are grayscale, centered, and rendered on clean backgrounds, while SVHN digits are RGB, embedded in cluttered street-scene photographs, and exhibit substantial intra-class variation in font, scale, and orientation. The per-channel pixel mean alone quantifies the gap—0.130 for MNIST versus 0.451 for SVHN—and a classifier trained exclusively on MNIST source samples achieves only 23.27% accuracy on the SVHN test set in our experiments, barely above the 10% random-chance baseline for a ten-class problem.

Unsupervised domain adaptation (UDA) addresses covariate shift by aligning the source and target feature distributions during training, without access to target labels [Ganin et al., 2016]. The Wasserstein distance, grounded in optimal transport (OT) theory, provides a geometrically principled measure of the discrepancy between two probability distributions and has been applied successfully as a domain alignment objective in prior work [Courty et al., 2015]. However, computing the exact Wasserstein distance between two empirical distributions of size n requires solving a linear program with $\mathcal{O}(n^2)$ variables, or at best running the Sinkhorn algorithm [Cuturi, 2013] on an $n \times n$ cost matrix, an $\mathcal{O}(n^2)$ memory footprint that grows prohibitively with batch size. In our benchmarks, Sinkhorn at $n = 5,000$ takes 40,542 ms per call versus 48 ms for SWD at $L = 100$ projections: an $838\times$ slowdown that renders full OT impractical for use as a per-batch training loss in any realistic deep learning pipeline.

Sliced Wasserstein Distance (SWD) [Rabin et al., 2011] addresses this scalability limitation by replacing the intractable d -dimensional OT problem with a Monte Carlo average of one-dimensional Wasserstein distances along random projection directions $\theta \in S^{d-1}$. Each one-dimensional subproblem reduces to sorting, $\mathcal{O}(n \log n)$, and the resulting estimator is a valid metric on probability distributions [Bonneel et al., 2014]. Max-Sliced Wasserstein Distance (Max-SWD) [Deshpande et al., 2019] further improves on SWD by replacing the average over random projections with an adversarial maximization: rather than diluting the alignment signal across uninformative directions, Max-SWD identifies the single projection direction $\theta^* \in S^{d-1}$ that maximizes the one-dimensional Wasserstein distance between the two projected distributions. This paper makes the following contributions:

- We implement SWD, Max-SWD, and log-domain Sinkhorn from scratch in PyTorch, verify each implementation against closed-form results and reference libraries, and deploy all three as plug-in alignment losses in a shared convolutional encoder trained on the MNIST→SVHN UDA benchmark.
- We conduct a systematic empirical comparison of all three alignment losses against an unadapted baseline, reporting best and final target accuracy, per-epoch wall-clock time, and alignment loss dynamics over 20 training epochs.
- We derive and apply a $\sqrt{d}$ scaling correction for SWD and Max-SWD in L2-normalized embedding spaces, and demonstrate empirically that this correction is necessary for the alignment loss to produce a meaningful gradient signal relative to the cross-entropy loss in $d = 128$ dimensions.
- We run a projection ablation over $L \in \{10, 50, 100, 500, 1000, 2000, 5000\}$ at $d \in \{8, 32, 128\}$, establishing $L = 1,000$ as the practical convergence threshold for the SWD estimator at the embedding dimension used in training, and confirm that the $\mathcal{O}(1/\sqrt{L})$ convergence rate predicted by the central limit theorem holds empirically.

2 Background

2.1 Optimal Transport and the Wasserstein Distance

Optimal transport theory concerns the problem of finding the most efficient way to transport one probability measure into another, where efficiency is measured by a prescribed cost function [Fernandes Montesuma et al., 2023]. Given two probability measures μ and ν on $\mathbb{R}^d$, the squared 2-Wasserstein distance is defined as the solution to the Kantorovich primal problem [Arjovsky et al., 2017]:

$$W_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 d\pi(x, y) \quad (1)$$

where $\Pi(\mu, \nu)$ denotes the set of all joint probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν , respectively. The transport plan $\pi(x, y)$ specifies the amount of mass transported from location x under μ to location y under ν , subject to the marginal constraints:

$$\int_{\mathbb{R}^d} d\pi(x, y) = d\nu(y), \quad \int_{\mathbb{R}^d} d\pi(x, y) = d\mu(x) \quad (2)$$

Unlike divergence-based measures such as the KL divergence or total variation distance, W_2^2 metrizes weak convergence of probability measures and is sensitive to the geometry of the underlying space [Arjovsky et al., 2017], making it particularly well-suited for comparing distributions supported on different regions of $\mathbb{R}^d$: a property directly relevant to domain adaptation, where source and target distributions are structurally disjoint in both pixel and feature space.

2.2 One-Dimensional Wasserstein Distance

In one dimension, the Kantorovich problem admits a closed-form solution that bypasses the linear program entirely. For two probability measures μ and ν on $\mathbb{R}$ with cumulative distribution functions F_μ and F_ν , the squared 2-Wasserstein distance reduces to the L^2 distance between their quantile functions [Cuturi, 2013]:

$$W_2^2(\mu, \nu) = \int_0^1 (F_\mu^{-1}(t) - F_\nu^{-1}(t))^2 dt \quad (3)$$

where $F^{-1}(t) = \inf\{x : F(x) \geq t\}$ is the quantile function of the respective measure. The optimal transport plan in one dimension is the monotone rearrangement: the i -th order statistic of μ is transported to the i -th order statistic of ν [Cuturi, 2013]. For two empirical distributions with n samples each, this reduces to:

$$W_2^2(\hat{\mu}_n, \hat{\nu}_n) = \frac{1}{n} \sum_{i=1}^n (x_{(i)} - y_{(i)})^2 \quad (4)$$

where $x_{(1)} \leq \dots \leq x_{(n)}$ and $y_{(1)} \leq \dots \leq y_{(n)}$ are the order statistics of the two samples. The computational cost is $\mathcal{O}(n \log n)$, dominated by the two sorting operations. We verified this implementation against the POT library [Flamary et al., 2021] on 500-sample Gaussian distributions, obtaining an absolute error of 1.23×10^{-6} .

2.3 Sliced Wasserstein Distance

The Sliced Wasserstein Distance [Rabin et al., 2011] extends the one-dimensional closed-form solution to arbitrary dimensions by integrating over all possible projection directions on the unit sphere. For a unit vector $\theta \in S^{d-1}$, the push-forward $\theta_{\#}\mu$ denotes the one-dimensional measure obtained by projecting μ onto the line defined by θ :

$$(\theta_{\#}\mu)(B) = \mu(\{x \in \mathbb{R}^d : \langle x, \theta \rangle \in B\}), \quad B \subseteq \mathbb{R} \quad (5)$$

The Sliced Wasserstein Distance is then defined as the expectation of the one-dimensional W_2^2 over directions drawn uniformly from S^{d-1} :

$$\text{SWD}_2^2(\mu, \nu) = \int_{S^{d-1}} W_2^2(\theta_{\#}\mu, \theta_{\#}\nu) d\sigma(\theta) \quad (6)$$

where σ denotes the uniform (Haar) measure on S^{d-1} . The SWD is a valid metric on the space of probability measures with finite second moments [Bonneel et al., 2014]—it is non-negative, equals zero if and only if $\mu = \nu$, is symmetric, and satisfies the triangle inequality—the last of which we verified numerically across 1,000 random distribution triplets in $\mathbb{R}^8$ with zero violations at $L = 500$ projections. In practice, the integral over S^{d-1} is approximated by a Monte Carlo estimator using L random unit directions $\{\theta_l\}_{l=1}^L$, sampled by normalizing i.i.d. Gaussian draws:

$$\widehat{\text{SWD}}_L(\mu, \nu) = \frac{1}{L} \sum_{l=1}^L W_2^2(\theta_{l\#}\mu, \theta_{l\#}\nu) \quad (7)$$

By the law of large numbers, $\widehat{\text{SWD}}_L \rightarrow \text{SWD}$ as $L \rightarrow \infty$, with convergence rate $\mathcal{O}(1/\sqrt{L})$ by the central limit theorem.

2.4 Max-Sliced Wasserstein Distance

A limitation of the average SWD estimator in high dimensions is that most random projection directions are approximately orthogonal to the true axis of distributional divergence, diluting the alignment signal [Deshpande et al., 2019]. Max-Sliced Wasserstein Distance [REFERENCE REQUIRED] addresses this by replacing the average over random projections with an adversarial maximization over the unit sphere:

$$\text{Max-SWD}(\mu, \nu) = \max_{\theta \in S^{d-1}} W_2^2(\theta_{\#}\mu, \theta_{\#}\nu) \quad (8)$$

Since $W_2^2(\theta_{\#}\mu, \theta_{\#}\nu)$ is continuous in θ on the compact set S^{d-1} , the maximum is always attained. By construction, $\text{Max-SWD}(\mu, \nu) \geq \text{SWD}(\mu, \nu)$, since the maximum over a set is at least as large as its average. The optimal direction θ^* is the projection that maximally separates the two distributions—analogue to the optimal discriminator in a generative adversarial network [Goodfellow et al., 2014]—and is found via projected gradient ascent on S^{d-1} :

$$\tilde{\theta}_{t+1} = \theta_t + \eta \cdot \nabla_{\theta} W_2^2(\theta_{\#}\hat{\mu}_n, \theta_{\#}\hat{\nu}_n) \Big|_{\theta_t}, \quad \theta_{t+1} = \frac{\tilde{\theta}_{t+1}}{\|\tilde{\theta}_{t+1}\|} \quad (9)$$

In a controlled experiment with $d = 8$ and distributional separation confined to dimension 0, projected gradient ascent recovered the true separation direction with $|\theta^* \cdot e_0| = 0.9990$ at convergence (100 steps, $\eta = 0.01$), while the average SWD at $L = 2,000$ underestimated the true gap by a factor of approximately 8 (0.48 versus 4.03).

3 Method

3.1 Domain Adaptation Objective

Let $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ denote the labeled source dataset (MNIST) and $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$ denote the unlabeled target dataset (SVHN). A shared encoder $f_{\theta} : \mathbb{R}^{3 \times 32 \times 32} \rightarrow S^{d-1}$ maps images from both domains into a common d -dimensional embedding space, and a classifier head $g_{\phi} : S^{d-1} \rightarrow \mathbb{R}^K$ maps embeddings to class logits. The total training loss combines a supervised source classification term with an unsupervised distributional alignment term:

$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{\text{CE}}(g_{\phi}(f_{\theta}(X_s)), Y_s) + \lambda \cdot D(f_{\theta}^{\text{align}}(X_s), f_{\theta}^{\text{align}}(X_t)) \quad (10)$$

where X_s, X_t are source and target mini-batches of size $n = 256$, Y_s are the corresponding source labels, f_θ^{align} denotes the alignment head path of the encoder (described in Section 3.3), D is the chosen alignment distance, and $\lambda > 0$ controls the relative weight of the alignment term. The cross-entropy loss uses label smoothing with $\alpha = 0.1$ [Szegedy et al., 2015]:

$$\mathcal{L}_{\text{CE}}(z, y) = -(1 - \alpha) \log p_y(z) - \frac{\alpha}{K} \sum_{k=1}^K \log p_k(z) \quad (11)$$

where $p_k(z) = \text{softmax}(g_\phi(z))_k$ and $K = 10$. Label smoothing prevents the classifier from becoming overconfident on source samples, which would suppress the cross-entropy gradient and allow the alignment term to dominate the combined loss [Szegedy et al., 2015]. A diagnostic epoch was run before the full training to verify that $\lambda \cdot D / \mathcal{L}_{\text{CE}} < 0.5$ for all methods, confirming that the alignment term does not overwhelm source classification in early training.

3.2 Alignment Loss Variants

Sliced Wasserstein Distance. The SWD alignment loss projects the source and target embedding batches onto $L = 200$ random unit directions and averages the one-dimensional W_2^2 distances. A $\sqrt{d}$ scaling correction is applied to all projected values to compensate for the concentration of projections in high dimensions: for L2-normalized embeddings on S^{d-1} , the expected squared projection onto a random unit direction satisfies $\mathbb{E}_\theta[\langle z, \theta \rangle^2] = 1/d$, causing the raw alignment loss to shrink by a factor of d relative to its unnormalized counterpart. The corrected estimator is:

$$\hat{\mathcal{L}}_{\text{SWD}} = \frac{d}{n \cdot L} \sum_{l=1}^L \sum_{i=1}^n \left(\langle x_{(i)}^l, \theta_l \rangle - \langle y_{(i)}^l, \theta_l \rangle \right)^2 \quad (12)$$

where $\langle x_{(i)}^l, \theta_l \rangle$ denotes the i -th order statistic of the source projections along direction θ_l , and analogously for target projections. The method-specific regularization weight is $\lambda_{\text{SWD}} = 0.08$.

Max-Sliced Wasserstein Distance. The Max-SWD alignment loss finds the worst-case projection direction via projected gradient ascent initialized at a random point on S^{d-1} . To keep per-batch overhead minimal, only 10 ascent steps are used with learning rate $\eta = 0.05$, compared to the 100 steps required for full convergence in the standalone setting. Gradient ascent is performed on detached source and target embeddings to prevent second-order gradient flow through the ascent loop; the final alignment loss is then computed with gradients enabled, allowing the encoder to receive a clean first-order signal. The same $\sqrt{d}$ scaling correction is applied. The method-specific regularization weight is $\lambda_{\text{Max-SWD}} = 0.04$.

Sinkhorn (Entropic OT Baseline). The Sinkhorn alignment loss computes the entropic OT distance between source and target embedding batches using the log-domain Sinkhorn-Knopp algorithm Cuturi [2013] with regularization parameter $\varepsilon = 1.0$ and 50 iterations. The larger ε value was chosen over the more common $\varepsilon = 0.1$ to ensure numerical stability in the log-domain updates during training; smaller values sharpen the transport plan but require significantly more iterations to converge and produce larger gradient magnitudes that risk destabilizing the encoder. No $\sqrt{d}$ correction is applied, as the Sinkhorn cost is computed directly from the squared Euclidean distance matrix in embedding space. The method-specific regularization weight is $\lambda_{\text{Sinkhorn}} = 0.05$.

3.3 Encoder Architecture

The encoder f_θ is a small convolutional network that processes images from both domains identically after unified preprocessing to $3 \times 32 \times 32$ tensors normalized to $[-1, 1]$. The architecture consists of three convolutional blocks, each comprising a 3×3 convolution, batch normalization [Ioffe and Szegedy, 2015], ReLU activation, and 2×2 max pooling, with channel widths $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$. The convolutional output is flattened and projected to $\mathbb{R}^d$ via a fully connected layer followed by ReLU, then L2-normalized to produce unit-norm embeddings on S^{d-1} :

$$f_\theta(x) = \frac{h_\theta(x)}{\|h_\theta(x)\|_2} \quad (13)$$

The encoder has two forward paths. The primary path `forward(x)` is used for source classification and passes through the main fully connected branch. The alignment path `forward_align(x)` is used for alignment loss computation and passes through a dedicated `align_head` branch—a separate linear projection followed by batch normalization, ReLU, and L2 normalization—for SWD and Max-SWD. This separation prevents alignment gradients from directly modifying the classification representation, stabilizing the cross-entropy loss throughout training. The classifier head g_ϕ is a single linear

layer mapping $\mathbb{R}^d \rightarrow \mathbb{R}^{10}$, trained exclusively on labeled source samples. The total parameter count is 357,258, of which 355,968 belong to the encoder and 1,290 to the classifier head.

3.4 Training Protocol

All four experimental conditions—no adaptation, SWD, Max-SWD, and Sinkhorn—use identical optimization settings to ensure a fair comparison. Parameters are updated with the Adam optimizer [Kingma and Ba, 2017] at learning rate 3×10^{-4} and weight decay 10^{-3} , with a cosine annealing learning rate schedule over $T = 20$ epochs [Loshchilov and Hutter, 2016]. Gradients are clipped to $\|\nabla\|_2 \leq 5.0$ before each parameter update. Both data loaders use batch size 256 with `drop_last=True` to guarantee fixed-size batches for the alignment loss. All experiments are run with global random seed 42 and full CUDA determinism (`cuda.deterministic=True`, `cuda.benchmark=False`) to ensure reproducibility. The complete set of hyperparameters is summarized in Table 1.

Table 1: Training hyperparameters shared across all experimental conditions.

Hyperparameter	Value
Optimizer	Adam
Learning rate	3×10^{-4}
Weight decay	10^{-3}
LR schedule	CosineAnnealingLR ($T_{\max} = 20$)
Epochs	20
Batch size	256
Label smoothing α	0.1
Gradient clip (max norm)	5.0
Embedding dimension d	128
Random seed	42

4 Experiments

4.1 Datasets and Preprocessing

MNIST. The MNIST dataset [Lecun et al., 1998] consists of 60,000 training and 10,000 test grayscale images of handwritten digits, each of size $1 \times 28 \times 28$, spanning ten classes (digits 0–9). For compatibility with the shared encoder, MNIST images are resized to 32×32 , replicated across three channels, and normalized to $[-1, 1]$ per channel with mean and standard deviation 0.5.

SVHN. The Street View House Numbers (SVHN) dataset [Netzer et al., 2011] consists of 73,257 training and 26,032 test RGB images of house number digits cropped from Google Street View photographs, each of size $3 \times 32 \times 32$, spanning the same ten digit classes. SVHN images are normalized to $[-1, 1]$ per channel with mean and standard deviation 0.5. No target labels are used at any point during training.

Domain gap. The distributional gap between the two domains is substantial even at the pixel level. The per-channel mean pixel intensity is 0.130 for MNIST (replicated to three channels) versus 0.451, 0.457, and 0.485 for the R, G, and B channels of SVHN respectively—a factor of approximately $3.5\times$ in mean intensity. The per-channel standard deviation is 0.308 for MNIST versus approximately 0.204 for SVHN, reflecting MNIST’s bimodal pixel distribution (background versus digit stroke) versus SVHN’s smoother natural image statistics. Beyond pixel statistics, the two domains differ structurally: MNIST digits are centered, clean, and rendered on uniform backgrounds, while SVHN digits are embedded in cluttered natural scenes with variable font, scale, and orientation.

4.2 Evaluation Protocol

Target domain accuracy is evaluated on the full SVHN test set (26,032 samples) after every training epoch using the shared encoder and classifier head, without access to any target labels. No target data is used for model selection: the best accuracy reported for each method is the maximum across all 20 epochs, reflecting the peak alignment achieved during training.

Final accuracy refers to the accuracy at epoch 20. Wall-clock time per epoch is measured as the mean over all 20 epochs, including both the forward pass, alignment loss computation, backward pass, and optimizer step.

4.3 Benchmark Results

Table 2 reports the best target accuracy, final target accuracy, and mean wall-clock time per epoch for all four experimental conditions. All three alignment methods outperform the no-adaptation baseline on best accuracy, confirming that distributional alignment in feature space transfers discriminative structure from the source domain to the target domain even in the absence of target labels.

Table 2: Benchmark results on the MNIST→SVHN UDA task. Best accuracy is the maximum over all 20 epochs; final accuracy is the accuracy at epoch 20. Time per epoch is averaged over all 20 epochs. Bold indicates the best result in each column.

Method	Best Acc.	Final Acc.	Time / Epoch	$\lambda \cdot \text{Align (avg)}$
No Adaptation	23.27%	11.59%	19.6s	—
SWD ($L = 200$)	24.60%	21.08%	19.9s	0.00058
Max-SWD	28.19%	24.61%	21.0s	0.00144
Sinkhorn ($\varepsilon = 1.0$)	27.07%	18.48%	26.3s	0.03994

Max-SWD achieves the highest best accuracy at 28.19%, a 4.92 percentage point improvement over the no-adaptation baseline, at an average epoch time of 21.0 seconds: only 1.4 seconds above the baseline. SWD at $L = 200$ projections delivers a more modest gain of 1.33 percentage points at effectively zero additional cost (19.9s versus 19.6s), confirming that even a low-variance random projection estimator provides sufficient alignment signal to prevent source-domain collapse. Sinkhorn achieves an intermediate best accuracy of 27.07% but is the least stable method: it peaks at epoch 2—the same epoch as the no-adaptation baseline—and degrades continuously to a final accuracy of 18.48% by epoch 20, a drop of 8.6 percentage points from its peak. By contrast, SWD and Max-SWD both peak at epoch 9 and maintain stable accuracy plateaus thereafter, with drops of only 3.5 and 3.6 percentage points respectively, demonstrating that slicing-based alignment losses provide a sustained regularization signal that prevents the encoder from overfitting the source domain.

The gap between best accuracy and final accuracy is particularly informative as a stability metric. The no-adaptation baseline collapses from 23.27% at epoch 2 to 11.59% by epoch 20—a drop of 11.7 percentage points—as the encoder progressively specializes to MNIST statistics without any target domain constraint. Sinkhorn’s drop of 8.6 percentage points is nearly as severe, indicating that at $\varepsilon = 1.0$, the over-smoothed transport plan fails to provide a sustained alignment gradient beyond the first few epochs. The average alignment term $\lambda \cdot \text{align} = 0.03994$ for Sinkhorn is two orders of magnitude larger than that of SWD (0.00058) and Max-SWD (0.00144), yet produces worse final accuracy—a direct consequence of the diffuse, unfocused gradient produced by an over-regularized transport plan.

4.4 Projection Ablation

To characterize the convergence behavior of the SWD Monte Carlo estimator and determine the minimum number of projections required for a reliable estimate at the embedding dimension used in training, we ran an ablation over $L \in \{10, 50, 100, 500, 1000, 2000, 5000\}$ at $d = 128$, using $n = 500$ samples and 20 independent trials per setting. The coefficient of variation ($\text{CV} = \text{std}/\text{mean}$) was used as the convergence criterion, with a practical threshold of $\text{CV} < 5\%$.

Table 3: SWD projection ablation at $d = 128$, $n = 500$ samples, 20 independent trials per L . $\text{CV} = \text{std}/\text{mean}$. Converged indicates $\text{CV} < 5\%$.

L	Mean SWD	Std	CV	Time (ms)	Converged
10	0.2464	0.0944	0.3833	1.10	No
50	0.2674	0.0367	0.1371	3.45	No
100	0.2547	0.0286	0.1124	3.90	No
500	0.2505	0.0131	0.0522	17.38	No
1000	0.2520	0.0099	0.0395	35.56	Yes
2000	0.2537	0.0090	0.0355	75.28	Yes
5000	0.2564	0.0040	0.0156	211.62	Yes

The CV decays as $\mathcal{O}(1/\sqrt{L})$, consistent with the Monte Carlo central limit theorem: from $L = 10$ to $L = 5,000$, the CV reduces by a factor of approximately 24.6, against the theoretical prediction of $\sqrt{5000/10} \approx 22.4$. The practical convergence threshold of $\text{CV} < 5\%$ is first reached at $L = 1,000$ ($\text{CV} = 3.95\%$, 35.56 ms), confirming $L = 1,000$ as the recommended setting for reliable SWD estimation at $d = 128$. Notably, this convergence threshold is reached at a wall-clock cost of 35.56 ms—approximately $33\times$ faster than Sinkhorn at equivalent sample size ($n = 1,000$: 1,190 ms)—demonstrating that even the high-precision SWD estimator is substantially more efficient than full entropic OT.

The same ablation was conducted at $d \in \{8, 32\}$ to assess the effect of embedding dimensionality on estimator convergence. The minimum L achieving $\text{CV} < 5\%$ is 500 at $d = 8$ ($\text{CV} = 4.85\%$) and 1,000 at $d = 32$ ($\text{CV} = 3.39\%$) and $d = 128$ ($\text{CV} = 3.95\%$). The modest increase in required L from $d = 8$ to $d = 32$ and $d = 128$ —a single doubling of the convergence threshold despite a $16\times$ increase in dimensionality—is consistent with a well-known geometric property of high-dimensional spheres: as d grows, random projections become increasingly uniform in their coverage of S^{d-1} , partially compensating for the larger space being averaged over.

5 Discussion

5.1 Accuracy-Speed Tradeoff

The benchmark results establish a clear Pareto frontier across the three alignment methods. Max-SWD occupies the optimal position: it achieves the highest best accuracy (28.19%), the highest final accuracy (24.61%), and the most stable training dynamics, at an epoch time of 21.0 seconds, only 7% above the no-adaptation baseline. SWD at $L = 200$ projections occupies the speed-optimal position: it adds less than 2% per-epoch overhead while delivering consistent alignment that prevents the source-domain collapse observed in the unadapted baseline, making it the natural choice when training time is the binding constraint. Sinkhorn is strictly Pareto-dominated by Max-SWD: it incurs the largest time penalty (26.3 seconds per epoch, 34% above baseline) while producing the least stable alignment dynamics, with a final accuracy of 18.48% that is lower than the no-adaptation best accuracy of 23.27%. The practical recommendation is unambiguous: for feature-space domain adaptation with L2-normalized embeddings, Max-SWD should be preferred over both SWD and Sinkhorn whenever a modest computational budget is available, and SWD should be preferred over Sinkhorn in all settings where training stability matters.

5.2 What Max-SWD Captures That SWD Misses

The performance gap between Max-SWD (28.19%) and SWD (24.60%) has a precise geometric explanation rooted in the structure of the domain gap in embedding space. When the distributional divergence between source and target is concentrated along a small number of directions—as is typical when both domains share the same label space and differ primarily in appearance [Ganin et al., 2016]—averaging over L random projection directions dilutes the alignment signal. Specifically, if the true divergence is concentrated in a d_{gap} -dimensional subspace of $\mathbb{R}^d$, the expected fraction of random projections that land in this subspace is approximately d_{gap}/d . For $d = 128$ and a domain gap that is plausibly low-dimensional, this means that the vast majority of the $L = 200$ projections used in SWD training contribute near-zero signal to the alignment gradient, while the few informative projections are averaged away. Max-SWD eliminates this dilution by construction: the projected gradient ascent converges to θ^* , the direction of maximum distributional divergence, and the entire alignment gradient is concentrated along this single direction. This is precisely the mechanism by which Max-SWD achieves a 3.59 percentage point advantage over SWD despite using a single effective projection rather than 200, and why the adversarial formulation of Max-SWD is not merely a theoretical refinement but a practically consequential one.

5.3 Limitations

Several limitations of the present study should be acknowledged. First, all results are reported from a single random seed (42), and the accuracy differences between methods—particularly between SWD (24.60%) and Sinkhorn (27.07%) on best accuracy—may not be statistically significant without multi-seed evaluation and confidence intervals. Second, the absolute accuracy levels achieved (23–28%) reflect the intrinsic difficulty of the MNIST→SVHN benchmark under purely unsupervised alignment: without any target supervision, the encoder has no direct feedback on whether the aligned representations are class-discriminative in the target domain, and the alignment loss alone cannot guarantee that source class boundaries transfer correctly to the target domain Ben-David et al. [2010]. Third, the Sinkhorn regularization parameter $\varepsilon = 1.0$ was chosen for numerical stability rather than through systematic hyperparameter search; a smaller ε might yield a sharper transport plan and more sustained alignment signal, at the cost of slower convergence and potential instability. Fourth, the λ values for each method were selected via a single diagnostic epoch and not through cross-validation; a more rigorous hyperparameter search might narrow the performance gap between methods. Fifth, the projection ablation was

conducted on synthetic Gaussian distributions rather than on the actual encoder embeddings produced during training, and the convergence threshold $L = 1,000$ derived from this ablation may not precisely characterize the variance of the SWD estimator on the true source-target embedding distributions encountered in the training loop.

5.4 Connections to Prior Work

The domain adaptation framework implemented in this paper is most closely related to approaches that minimize an integral probability metric between source and target feature distributions as a regularization term during classifier training [Ganin et al., 2016]. The use of Wasserstein distance as the alignment objective connects this work to Wasserstein-distance-based domain adaptation methods [Courty et al., 2015], which have demonstrated strong empirical performance on standard benchmarks but typically rely on full OT or its dual formulation via neural network discriminators [Arjovsky et al., 2017]. The slicing approach adopted here trades the exactness of full OT for scalability, following the line of work initiated by Rabin et al. [Rabin et al., 2011] and extended by Kolouri et al. [Kolouri et al., 2018] to generative modeling and distribution matching. The adversarial projection in Max-SWD connects to the broader literature on adversarial training [Goodfellow et al., 2014] and to the specific line of work on projection-based adversarial domain adaptation [Deshpande et al., 2019]. The $\sqrt{d}$ scaling correction for L2-normalized embeddings, to our knowledge, has not been explicitly identified in prior SWD literature as a necessary implementation detail for high-dimensional normalized embedding spaces, and represents a practically important contribution of this work.

6 Conclusion

This paper presented a systematic empirical study of Sliced Wasserstein Distance and Max-Sliced Wasserstein Distance as scalable alignment losses for unsupervised domain adaptation on the MNIST→SVHN benchmark. All three alignment distances—SWD, Max-SWD, and full entropic OT via Sinkhorn—were implemented from scratch in PyTorch, verified against closed-form results and reference libraries, and evaluated under identical training conditions against an unadapted baseline. The central finding is that Max-SWD achieves the best accuracy-speed tradeoff among all methods tested: its adversarial projection mechanism concentrates the alignment gradient onto the single most informative direction in embedding space, yielding a 4.92 percentage point improvement over the no-adaptation baseline at only 7% additional wall-clock overhead per epoch. SWD at $L = 200$ projections provides a complementary operating point—negligible computational overhead at a modest accuracy gain—making it the natural choice when training time is the primary constraint. Sinkhorn, despite its theoretical status as the gold-standard OT distance, is strictly Pareto-dominated by Max-SWD on every practical metric: it is 34% slower per epoch, produces unstable alignment dynamics that peak transiently and degrade continuously, and finishes with a final accuracy lower than the no-adaptation best accuracy.

Two implementation findings of independent practical interest emerged from this study. First, a $\sqrt{d}$ scaling correction is necessary for SWD and Max-SWD to function as effective alignment losses in L2-normalized embedding spaces: without it, the alignment loss is negligibly small relative to the cross-entropy loss in $d = 128$ dimensions, and the encoder receives no meaningful domain alignment gradient. Second, the projection ablation establishes $L = 1,000$ as the practical convergence threshold for the SWD Monte Carlo estimator at $d = 128$ —the point at which the coefficient of variation drops below 5%—at a wall-clock cost of 35.56 ms, approximately 33× faster than Sinkhorn at equivalent sample size. The $\mathcal{O}(1/\sqrt{L})$ convergence rate predicted by the central limit theorem was confirmed empirically, with the observed reduction factor of 24.6 from $L = 10$ to $L = 5,000$ in close agreement with the theoretical prediction of 22.4.

Several directions remain open for future investigation. A multi-seed evaluation with confidence intervals would establish whether the accuracy differences between methods are statistically significant and would quantify the variance introduced by the stochastic alignment gradient. Combining Max-SWD alignment with pseudo-label refinement [Lee, 2013] on high-confidence target predictions could close the remaining gap between unsupervised alignment and supervised target performance, as pseudo-labeling provides direct class-discriminative feedback that the alignment loss alone cannot supply. An adaptive λ schedule—increasing the alignment weight as the encoder converges and the cross-entropy loss stabilizes—could improve the sustained alignment signal of all three methods without requiring method-specific hyperparameter tuning. Finally, extending the projection ablation to the actual encoder embedding distributions encountered during training, rather than synthetic Gaussian distributions, would provide a more precise characterization of the SWD estimator variance in the domain adaptation setting and could inform tighter recommendations for the minimum L required in practice.

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